

Q find sum

a) $s = a^{100} + a^{99}b + a^{98}b^2 + \dots + ab^{99} + b^{100}$

b) $s = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots$ (upto n terms)

a) Soln first term = a^{100}

$$r = \frac{b}{a}$$

$$n = 101$$

$$S = a^{100} \frac{1 - \left(\frac{b}{a}\right)^{101}}{1 - \frac{b}{a}}$$

$$= a^{100} \left[\frac{1 - \frac{b^{101}}{a^{101}}}{1 - \frac{b}{a}} \right]$$

$$= a^{100} \times \frac{\frac{a^{101} - b^{101}}{a^{101}}}{\frac{a-b}{a}}$$

$$= \cancel{a^{100}} \times \frac{(a^{101} - b^{101}) \cancel{a}}{(a-b) \cancel{a^{101}}}$$

$$= \frac{a^{101} - b^{101}}{a-b}$$

$$S = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots \text{ (upto } n \text{ terms)}$$

$$a = 1, r = -\frac{1}{2}, n = n$$

$$S = 1 \cdot \frac{1 - \left(-\frac{1}{2}\right)^n}{1 - \left(-\frac{1}{2}\right)}$$

$$= \frac{1 - \left(-\frac{1}{2}\right)^n}{\left(\frac{3}{2}\right)}$$

$$= \left(\frac{2}{3}\right) \left[1 - \left(-\frac{1}{2}\right)^n\right]$$

$$\frac{-\frac{1}{2}}{1} = -\frac{1}{2}$$

$$\frac{+\frac{1}{4}}{-\frac{1}{2}} = \frac{-2}{4} = -\frac{1}{2}$$

$$S = 0.6 + 0.66 + 0.666 + \dots \text{ (upto } n \text{ terms)}$$

$$= 6 [0.1 + 0.11 + 0.111 + \dots]$$

$$= \frac{6}{9} [0.9 + 0.99 + 0.999 + \dots]$$

$$= \frac{2}{3} [(1-0.1) + (1-0.01) + (1-0.001) + \dots]$$

$$= \frac{2}{3} \left[(1+1+1+\dots) - \underbrace{(0.1 + 0.01 + 0.001 + \dots)}_{a=0.1, r=\frac{1}{10}, n \text{ terms}} \right]$$

$$= \frac{2}{3} \left[n - 0.1 \frac{1 - \left(\frac{1}{10}\right)^n}{1 - \frac{1}{10}} \right]$$

$$= \frac{2}{3} \left[n - \frac{1}{10} \times \frac{10}{9} \left(1 - \frac{1}{10^n}\right) \right]$$

$$= \frac{2}{3} \left[n - \frac{1}{9} \left(1 - \frac{1}{10^n}\right) \right]$$

$$\frac{0.66}{0.6} = \frac{66}{60} = \frac{11}{10}$$

$$\frac{0.666}{.66} = \frac{666}{660} = \frac{111}{110}$$

$$a \frac{(1-r^n)}{1-r}$$

$$S = 7 + 77 + 777 + \dots \quad (\text{upto } n \text{ terms})$$

$$= 7 (1 + 11 + 111 + 1111 + \dots)$$

$$= \frac{7}{9} (9 + 99 + 999 + \dots)$$

$$= \frac{7}{9} ((10-1) + (100-1) + (1000-1) + \dots)$$

$$= \frac{7}{9} \left[(10 + 100 + 1000 + \dots) - (1 + 1 + 1 + \dots) \right]$$

$a=10, r=10, n=n$

$$= \frac{7}{9} \left[10 \times \frac{1-10^n}{1-10} - n \right]$$

$$= \frac{7}{9} \left[10 \times \frac{(10^n-1)}{9} - n \right]$$

Q Determine the no of terms of the GP
whose $a=3$, $a_n=96$ and $S_n=189$.

$$a=3 \quad | \quad a_n=96$$

$$ar^{n-1}=96$$

$$3 \cdot r^{n-1}=96$$

$$r^{n-1}=32$$

$$\frac{r^n}{r}=32$$

$$r^n=32r$$

$$2^n=64$$

$$\therefore n=6$$

$$S_n=189$$

$$a \cdot \frac{1-r^n}{1-r}=189$$

$$\frac{1-r^n}{1-r}=63$$

$$1-r^n=63-63r$$

$$1-32r=63-63r$$

$$31r=62$$

$$r=2$$

Q The sum of certain **infinite** GP is 20, When all the terms are in series are squared, the sum of resulting series is 80. Then find The first term of original series.

Soln

original GP $\rightarrow a, ar, ar^2, ar^3, \dots$ $\curvearrowright S = 20$

$$\frac{a}{1-r} = 20$$

$$a = 20(1-r)$$

Squared GP $\rightarrow a^2, a^2r^2, a^2r^4, a^2r^6, \dots$

$$a = 20 \left(1 - \frac{2}{3}\right) = \frac{20}{3}$$

$$S' = 80$$

$$\frac{a^2}{1-r^2} = 80$$

$$\frac{\cancel{20}^2 (1-r)^2}{(1-r^2)} = \cancel{80}$$

$$\frac{5(1-r)^2}{\cancel{(1-r)}(1+r)} = 1$$

$$\Rightarrow 5 - 5r = 1 + r$$

$$\Rightarrow 4 = 6r$$

$$\Rightarrow r = \frac{2}{3}$$

Properties of GP

① if all the terms of GP are multiplied or divided by a fixed number then resulting series is still a GP.

② $a_1, a_2, a_3, \dots, a_n \leftarrow \text{GP}$

$$a_1 a_n = a_2 a_{n-1} = a_3 a_{n-2} = \dots$$

③ 3 numbers in GP. $\Rightarrow \frac{a}{r}, a, ar$

4 numbers are in GP $\Rightarrow \frac{a}{r^3}, \frac{a}{r}, ar, ar^3$

AyP

$$1, 2 \cdot \underline{2}, 3 \cdot \underline{2^2}, 4 \cdot \underline{2^3}, 5 \cdot \underline{2^4}, 6 \cdot \underline{2^5}, \dots, 20 \cdot 2^{19}, \quad r = 2$$

$$1, \frac{3}{2}, \frac{5}{2^2}, \frac{7}{2^3}, \dots, \frac{19}{2^9} \quad \leadsto r = \frac{1}{2}$$

$$a, (a+d)r, (a+2d)r^2, \dots$$

sum of AP

infinite AP ($-1 < r < 1$)

$$r = -\frac{1}{3}$$

$$S = 1 + \frac{3}{3} + \frac{5}{3^2} + \frac{7}{3^3} + \dots$$

$$\frac{S}{3} = \frac{1}{3} + \frac{3}{3^2} + \frac{5}{3^3} + \frac{7}{3^4} + \dots$$

$$\frac{2S}{3} = 1 + \frac{2}{3} + \frac{2}{3^2} + \frac{2}{3^3} + \frac{2}{3^4} + \dots$$

$$a = \frac{2}{3}, r = \frac{1}{3}$$

$$= 1 + \frac{\binom{2}{3}}{1 - \binom{1}{3}}$$

$$\frac{2S}{3} = 2$$

$$S = 3$$

→ find sum of AP

$$a) S = 1 + \frac{2}{2} + \frac{3}{2^2} + \frac{4}{2^3} + \dots$$

$$b) S = 1 + 2x + 3x^2 + 4x^3 + \dots$$

$$c) S = 1 + 3x + 6x^2 + 10x^3 + \dots$$

$$c) S = 1 + 3x + 6x^2 + 10x^3 + \dots$$

$$x = x$$

$$Sx = x + 3x^2 + 6x^3 + 10x^4 + \dots$$

$$S - Sx = 1 + 2x + 3x^2 + 4x^3 + 5x^4 + \dots$$

$$(S - Sx)x = x + 2x^2 + 3x^3 + 4x^4 + \dots$$

$$= 1 + x + x^2 + x^3 + x^4 + \dots$$

$$a=1, r=x$$

$$S(1-x)^2 = \frac{1}{1-x}$$

$$S = \frac{1}{(1-x)^3}$$

$$(S - Sx) - (S - Sx)x$$

$$= (S - Sx)(1 - x)$$

$$= S(1-x)(1-x)$$

$$S = 1 + (1+x)y + (1+x+x^2)y^2 + (1+x+x^2+x^3)y^3 + \dots$$

$$x=y$$

$$Sy = y + (1+x)y^2 + (1+x+x^2)y^3 + \dots$$

$$S - Sy = 1 + xy + x^2y^2 + x^3y^3 + \dots$$

$$a=1, r=xy$$

$$S(1-y) = \frac{1}{1-xy}$$

$$S = \frac{1}{(1-xy)(1-y)}$$

Finite AP

$$S = 1 + 3 \cdot 3 + 5 \cdot 3^2 + 7 \cdot 3^3 + \dots$$

(upto 15 terms)

Soln

$$S = 1 + 3 \cdot 3 + 5 \cdot 3^2 + 7 \cdot 3^3 + \dots + 29 \cdot 3^{14}$$

$$1 + 14 \times 2 = 29$$

$$3S = 1 \cdot 3 + 3 \cdot 3^2 + 5 \cdot 3^3 + \dots + 27 \cdot 3^{14} + 29 \cdot 3^{15}$$

$$-2S = 1 + 2 \cdot 3 + 2 \cdot 3^2 + 2 \cdot 3^3 + \dots + 2 \cdot 3^{14} - 29 \cdot 3^{15}$$

$a = 6, r = 3, n = 14$

$$= 1 + \frac{(-3)}{6} \cdot \frac{1 - 3^{14}}{1 - 3} - 29 \cdot 3^{15}$$

$$-2S = 1 + 3(3^{14} - 1) - 29 \cdot 3^{15}$$

$$= 1 + 3^{15} - 3 - 29 \cdot 3^{15}$$

$$-2S = -2 - 28 \cdot 3^{15}$$

$$S = 1 + 14 \cdot 3^{15}$$

$$\rightarrow S = 1 + 2 \cdot 2 + 3 \cdot 2^2 + 4 \cdot 2^3 + \dots \quad (20 \text{ terms})$$

$$S = 1 + 2 \cdot 2 + 3 \cdot 2^2 + 4 \cdot 2^3 + \dots + 20 \cdot 2^{19}$$

$$2S = \quad 1 \cdot 2 + 2 \cdot 2^2 + 3 \cdot 2^3 + \dots + 19 \cdot 2^{19} + 20 \cdot 2^{20}$$

$$-S = 1 + 1 \cdot 2 + 1 \cdot 2^2 + 1 \cdot 2^3 + \dots + 1 \cdot 2^{19} - 20 \cdot 2^{20}$$

$a=1, r=2, n=20$

$$-S = 1 \cdot \frac{1-2^{20}}{1-2} - 20 \cdot 2^{20}$$

$$= 2^{20} - 1 - 20 \cdot 2^{20}$$

$$-S = -19 \cdot 2^{20} - 1$$

$$\underline{\underline{S = 19 \cdot 2^{20} + 1}}$$

Q If 1st, 2nd, 3rd term of an A.P. are 2, 6 and 16, find the 4th term of A.P.
(given common difference is negative)

$$a, (a+d)r, (a+2d)r^2, (a+3d)r^3$$

$$a = 2 \quad \left| \begin{array}{l} (a+d)r = 6 \\ (2+d)r = 6 \end{array} \right| \begin{array}{l} (a+2d)r^2 = 16 \\ (2+2d)r^2 = 16 \end{array}$$

$$r = \frac{6}{(2+d)}$$

$$\Rightarrow \frac{9(1+d)}{(2+d)^2} = 2$$

$$\Rightarrow 9+9d = 2d^2+8d+8$$

$$\Rightarrow 2d^2-d-1=0$$

$$\Rightarrow 2d^2-2d+d-1=0$$

$$\Rightarrow 2d(d-1)+1(d-1)=0$$

$$\Rightarrow (2d+1)(d-1)=0$$

$$d = -\frac{1}{2} \checkmark \quad \left| \quad d = 1 \quad \times \right.$$

$$r = \frac{6}{(2-\frac{1}{2})} = 4$$

$$a = 2$$

$$t_4 = (a+3d)r^3$$

$$= (2-\frac{3}{2})64$$

$$= 32$$

Harmonic Progression

HP $\rightarrow 1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \frac{1}{9}, \frac{1}{11}, \frac{1}{13}$

HP $\rightarrow 12, 6, 4, 3, \frac{12}{5}, 2$

$10, 8, \frac{20}{3}, \frac{40}{7}, \frac{40}{8}, \frac{40}{9}$

$\hookrightarrow 1, 3, 5, 7, 9$ AP $\checkmark$

$\hookrightarrow \frac{1}{12}, \frac{1}{6}, \frac{1}{4}, \frac{1}{3}, \frac{5}{12}, \frac{1}{2}$ AP $\checkmark$

$\hookrightarrow \frac{1}{12}, \frac{2}{12}, \frac{3}{12}, \frac{4}{12}, \frac{5}{12}, \frac{6}{12}$

$\hookrightarrow \frac{1}{10}, \frac{1}{8}, \frac{3}{20}, \frac{7}{40}$

$\hookrightarrow \frac{4}{40}, \frac{5}{40}, \frac{6}{40}, \frac{7}{40}, \frac{8}{40}, \frac{9}{40}$

$$\frac{1}{6} - \frac{1}{12} = \frac{2-1}{12} = \frac{1}{12}$$

$$\frac{1}{4} - \frac{1}{6} = \frac{3-2}{12} = \frac{1}{12}$$

$$\frac{1}{3} - \frac{1}{4} = \frac{4-3}{12} = \frac{1}{12}$$

$$\frac{1}{3} + \frac{1}{12} = \frac{5}{12}$$

$$\frac{5}{12} + \frac{1}{12} = \frac{6}{12}$$

$\rightarrow$ reciprocal of terms of an HP is AP and vice versa.

$$\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \frac{1}{a+3d}, \dots, \frac{1}{a+(n-1)d}$$

$$h_n = \frac{1}{a+(n-1)d}$$

Q If 5th and 8th term of a HP are 40 and 25 respectively, then find 10th term and 15th term of HP.

Q Let $a_1, a_2, a_3, \dots, a_{10}$ be an AP and $h_1, h_2, h_3, \dots, h_{10}$ be a HP and given $a_1 = 2 = h_1$ and $a_{10} = 3 = h_{10}$, then find value of

$$a_4 h_7 = ?$$

Q If $h_1, h_2, h_3, \dots, h_n$ be HP, then find value of

$$h_1 h_2 + h_2 h_3 + h_3 h_4 + \dots + h_{n-1} h_n = ?$$

$$h_5 = 40$$

$$\frac{1}{a+4d} = 40$$

$$a+4d = \frac{1}{40} \quad \text{--- (1)}$$

$$h_8 = 25$$

$$\frac{1}{a+7d} = 25$$

$$a+7d = \frac{1}{25} \quad \text{--- (2)}$$

$$h_{10}?, h_{15}?$$

$$(2) - (1)$$

$$(a+7d) - (a+4d) = \frac{1}{25} - \frac{1}{40}$$

$$3d = \frac{8-5}{200}$$

$$3d = \frac{3}{200}$$

$$d = \frac{1}{200}$$

$$d = \frac{1}{200} \text{ in (1)}$$

$$a + \frac{4}{200} = \frac{1}{40}$$

$$a = \frac{1}{40} - \frac{1}{50}$$

$$= \frac{5-4}{200}$$

$$= \frac{1}{200}$$

$$h_{10} = \frac{1}{a+9d}$$

$$= \frac{1}{\frac{1}{200} + \frac{9}{200}}$$

$$= \frac{1}{\frac{10}{200}}$$

$$= \frac{200}{10}$$

$$= 20$$

$$h_{15} = \frac{1}{a+14d}$$

$$= \frac{1}{\frac{1}{200} + \frac{14}{200}}$$

$$= \frac{1}{\frac{15}{200}}$$

$$= \frac{200}{15}$$

$$= \frac{40}{3}$$

Soln

$a_1, a_2, \dots, a_{10}$ AP

$h_1, h_2, \dots, h_{10} \leftarrow HP$

$$a_1 = 2$$

$$a = 2$$

$$a_{10} = 3$$

$$a + 9d = 3$$

$$2 + 9d = 3$$

$$9d = 1$$

$$d = \frac{1}{9}$$

$$a_4 = ?$$

$$a_4 = a + 3d$$

$$= 2 + \frac{1}{3}$$

$$= \frac{7}{3}$$

$$h_1 = 2$$

$$\frac{1}{a'} = 2$$

$$a' = \frac{1}{2}$$

$$h_{10} = 3$$

$$\frac{1}{a' + 9d'} = 3$$

$$a' + 9d' = \frac{1}{3}$$

$$\frac{1}{2} + 9d' = \frac{1}{3}$$

$$9d' = \frac{1}{3} - \frac{1}{2}$$

$$= \frac{2-3}{6}$$

$$9d' = -\frac{1}{6}$$

$$d' = -\frac{1}{54}$$

$$h_7 = \frac{1}{a' + 6d'}$$

$$= \frac{1}{\left(\frac{1}{2}\right) + \left(\frac{6}{54}\right)}$$

$$= \frac{1}{\frac{27+6}{54}}$$

$$= \frac{54}{21}$$

$$= \frac{18}{7}$$

$$a_4 h_7 = \frac{7}{3} \times \frac{18}{7} = 6$$

Q $h_1, h_2, h_3, \dots, h_n$ are in HP, then find
 $h_1 h_2 + h_2 h_3 + \dots + h_{n-1} h_n = ?$

Soln $h_1, h_2, h_3, \dots, h_n \rightarrow a_1, a_2, a_3, \dots, a_n \leftarrow \text{AP}$
 $\text{HP} \rightarrow \text{cd} = d$

$$= h_1 h_2 + h_2 h_3 + h_3 h_4 + \dots + h_{n-1} h_n$$

$$= \frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + \frac{1}{a_3 a_4} + \dots + \frac{1}{a_{n-1} a_n}$$

$$\frac{1}{d} \left[\frac{d}{a_1 a_2} + \frac{d}{a_2 a_3} + \frac{d}{a_3 a_4} + \dots + \frac{d}{a_{n-1} a_n} \right]$$

$$\frac{1}{d} \left[\frac{a_2 - a_1}{a_1 a_2} + \frac{a_3 - a_2}{a_2 a_3} + \frac{a_4 - a_3}{a_3 a_4} + \dots + \frac{a_n - a_{n-1}}{a_{n-1} a_n} \right]$$

$$\frac{1}{d} \left[\left(\frac{1}{a_1} - \frac{1}{a_2} \right) + \left(\frac{1}{a_2} - \frac{1}{a_3} \right) + \left(\frac{1}{a_3} - \frac{1}{a_4} \right) + \dots + \left(\frac{1}{a_{n-1}} - \frac{1}{a_n} \right) \right]$$

$$\frac{a_2 - a_1}{a_1 a_2}$$

$$\frac{a_2}{a_1 a_2} - \frac{a_1}{a_1 a_2}$$

$$\frac{1}{a_1} - \frac{1}{a_2}$$

$$= \frac{1}{d} \left[\frac{1}{a_1} - \frac{1}{a_n} \right]$$

$$= \frac{n-1}{\cancel{a_n - a_1}} \times \frac{\cancel{a_n - a_1}}{a_1 a_n}$$

$$= (n-1) \left(\frac{1}{a_1} \right) \left(\frac{1}{a_n} \right)$$

$$= (n-1) h_1 h_n$$

$$a_n = a_1 + (n-1)d$$

$$d = \frac{a_n - a_1}{n-1}$$